

C959

# Discrete Mathematics I

Armored Drill

8 concepts · 12 items · 4 transfer items

**What this is.** Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

**Source.** Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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# How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | _____  |                               |

# Items

1.

The IT department of a regional bank with 14 branches states: “If a server is patched, then it passes the audit.” The audit is held each March. Which statement is logically equivalent?

- A. Every server passes the audit.
- B. If a server is not patched, then it does not pass the audit.
- C. If a server passes the audit, then it is patched.
- D. If a server does not pass the audit, then it is not patched.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

2. [TRANSFER — reversed stem / different variable]

For which truth values is  $p \rightarrow q$  false?

- A. p false, q true
- B. p true, q false
- C. p false, q false
- D. Whenever q is false

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

3.

A compliance report at a hospital with 1,200 staff claims: “Every employee completed the 45-minute privacy training.” Which statement is the negation of this claim?

- A. No employee completed the privacy training.
- B. Every employee took longer than 45 minutes.
- C. Some employee completed the privacy training.
- D. At least one employee did not complete the privacy training.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

4.

A student council survey in October found that of 120 students, 70 take Python, 50 take Java, and 20 take both. How many take neither?

- A. 0
- B. 100
- C. 20
- D. 40

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

5. [TRANSFER — reversed stem / different variable]

$|A \cup B| = 90$ ,  $|A| = 60$ , and  $|B| = 45$ . What is  $|A \cap B|$ ?

- A. 105
- B. 30
- C. 15
- D. 45

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

6.

An engineer designing alarm logic for a building with 12 floors needs to simplify  $(x \cdot y)'$  before sending it to a supplier in Ohio. Which expression is equivalent?

- A.  $x' \cdot y'$
- B.  $x + y$
- C.  $x' + y'$
- D.  $x' + y$

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

7.

A theater renovated in 2020 has 15 rows. The first row has 20 seats and each row after has 2 more than the one before. Tickets cost \$45. How many seats are there in total?

- A. 48
- B. 300
- C. 720
- D. 510

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

8. [TRANSFER — reversed stem / different variable]

The series  $3 + 6 + 12 + \dots$  doubles each term. How many terms are needed for the sum to equal 381?

- A. 7
- B. 128
- C. 127
- D. 6

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

9.

At a company with 5 departments and 300 staff, relation R holds when employee a works in the same department as employee b. Everyone belongs to exactly one department. What kind of relation is R?

- A. A partial order
- B. An equivalence relation
- C. Symmetric only
- D. A function

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

10.

A contractor installed 10 routers in 2021, each costing \$900. Each router has a direct link to exactly 3 others. How many links are there?

- A. 15
- B. 45
- C. 9
- D. 30

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

11. [TRANSFER — reversed stem / different variable]

A graph has 12 edges and every vertex has degree 4. How many vertices does it have?

- A. 6
- B. 3
- C. 12
- D. 48

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

12.

A school district with 3 campuses connects 25 computers so that every pair can communicate, with no redundant loops. How many cables are needed?

- A. 300
- B. 24
- C. 25
- D. 3

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

# STOP

Answers and dissection follow. Finish every item and every strip first.

# Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

## 1. Answer D Concept 1

The IT department of a regional bank with 14 branches states: "If a server is patched, then it passes the audit." The audit is held each March. Which statement is logically equivalent?

**Target word:** equivalent form **Decides:** The contrapositive is equivalent.

|   |                    |                                                       |
|---|--------------------|-------------------------------------------------------|
| A | Absolute language  | The conditional says nothing about unpatched servers. |
| B | Inverted operation | The inverse, which is not equivalent.                 |
| C | Confusion set      | The converse, which is not equivalent.                |
| D | <b>Correct</b>     | $\neg q \rightarrow \neg p$ is the contrapositive.    |

## 2. Answer B Concept 1 • transfer

For which truth values is  $p \rightarrow q$  false?

**Target word:** the false row **Decides:** True premise, false conclusion.

|   |                    |                                             |
|---|--------------------|---------------------------------------------|
| A | Inverted operation | A false premise makes the conditional true. |
| B | <b>Correct</b>     | The only row that breaks the promise.       |
| C | Confusion set      | Also true: a false premise cannot break it. |
| D | Dropped element    | Ignores p; if p is also false it is true.   |

## 3. Answer D Concept 2

A compliance report at a hospital with 1,200 staff claims: "Every employee completed the 45-minute privacy training." Which statement is the negation of this claim?

**Target word:** negation of a universal **Decides:**  $\neg \forall$  becomes  $\exists \neg$ .

|   |                   |                                                  |
|---|-------------------|--------------------------------------------------|
| A | Absolute language | Too strong; negation needs only one exception.   |
| B | Camouflage        | Answers the time detail, which is not the claim. |
| C | Adjacent category | Consistent with the original; not its negation.  |
| D | <b>Correct</b>    | One counterexample makes "every" false.          |

## 4. Answer C Concept 3

A student council survey in October found that of 120 students, 70 take Python, 50 take Java, and 20 take both. How many take neither?

**Target word:** count outside both sets **Decides:**  $|P \cup J| = 70 + 50 - 20 = 100$ .

|   |                    |                                                    |
|---|--------------------|----------------------------------------------------|
| A | Dropped element    | Adds $70 + 50 = 120$ without removing the overlap. |
| B | Adjacent category  | That is how many take at least one.                |
| C | <b>Correct</b>     | $120 - 100 = 20$ .                                 |
| D | Inverted operation | Subtracts the overlap twice: $70 + 50 - 40 = 80$ . |

### 5. Answer C Concept 3 • transfer

$|A \cup B| = 90$ ,  $|A| = 60$ , and  $|B| = 45$ . What is  $|A \cap B|$ ?

**Target word:** solve for intersection **Decides:**  $60 + 45 - |A \cap B| = 90$ .

|   |                    |                                                   |
|---|--------------------|---------------------------------------------------|
| A | Inverted operation | Adds the sets instead of solving for the overlap. |
| B | Dropped element    | $90 - 60$ ignores B.                              |
| C | <b>Correct</b>     | $105 - 90 = 15$ .                                 |
| D | Wrong stem number  | The size of B.                                    |

### 6. Answer C Concept 4

An engineer designing alarm logic ~~for a building with 12 floors~~ needs to simplify  $(x \cdot y)'$  ~~before sending it to a supplier in Ohio~~. Which expression is equivalent?

**Target word:** apply De Morgan **Decides:** Complement each and swap AND to OR.

|   |                    |                                               |
|---|--------------------|-----------------------------------------------|
| A | Confusion set      | Keeps AND; that equals $(x + y)'$ .           |
| B | Inverted operation | Swaps the operator but drops the complements. |
| C | <b>Correct</b>     | $(xy)' = x' + y'$ .                           |
| D | Dropped element    | Complements only one variable.                |

### 7. Answer D Concept 5

A theater ~~renovated in 2020~~ has 15 rows. The first row has 20 seats and each row after has 2 more than the one before. ~~Tickets cost \$45~~. How many seats are there in total?

**Target word:** arithmetic series sum **Decides:** Last row  $= 20 + 14 \times 2 = 48$ ; sum  $= 15/2 \times (20 + 48)$ .

|   |                   |                                                 |
|---|-------------------|-------------------------------------------------|
| A | Confusion set     | That is the last row, not the total.            |
| B | Dropped element   | $20 \times 15$ ignores the increase.            |
| C | Adjacent category | $15 \times 48$ uses the last row for every row. |
| D | <b>Correct</b>    | $7.5 \times 68 = 510$ .                         |

### 8. Answer A Concept 5 • transfer

The series  $3 + 6 + 12 + \dots$  doubles each term. How many terms are needed for the sum to equal 381?

**Target word:** solve for n **Decides:**  $3(2^n - 1) = 381$ , so  $2^n = 128$ .

|   |                   |                                                                    |
|---|-------------------|--------------------------------------------------------------------|
| A | <b>Correct</b>    | $2^7 = 128$ .                                                      |
| B | Confusion set     | That is $2^n$ , not n.                                             |
| C | Dropped element   | Stops at $381 \div 3$ .                                            |
| D | Adjacent category | The exponent in the last term, $3 \times 2^6$ , which is $n - 1$ . |

### 9. Answer B Concept 6

At a company with 5 departments and 300 staff, relation R holds when employee a works in the same department as employee b. Everyone belongs to exactly one department. What kind of relation is R?

**Target word:** classify the relation **Decides:** Reflexive, symmetric, and transitive.

|          |                   |                                                                              |
|----------|-------------------|------------------------------------------------------------------------------|
| A        | Confusion set     | Needs antisymmetry; two coworkers relate both ways but are different people. |
| <b>B</b> | <b>Correct</b>    | It meets all three properties and splits staff into departments.             |
| C        | Dropped element   | Ignores that it is also reflexive and transitive.                            |
| D        | Adjacent category | Each employee relates to many coworkers, not one.                            |

### 10. Answer A Concept 7

A contractor installed 10 routers in 2021, each costing \$900. Each router has a direct link to exactly 3 others. How many links are there?

**Target word:** edge count **Decides:** Degree sum  $30 = 2 \times \text{edges}$ .

|          |                   |                                 |
|----------|-------------------|---------------------------------|
| <b>A</b> | <b>Correct</b>    | $30 \div 2 = 15$ .              |
| B        | Confusion set     | Complete graph on 10 vertices.  |
| C        | Adjacent category | Edges of a tree on 10 vertices. |
| D        | Dropped element   | Counts each link at both ends.  |

### 11. Answer A Concept 7 · transfer

A graph has 12 edges and every vertex has degree 4. How many vertices does it have?

**Target word:** solve for vertices **Decides:**  $4V = 2 \times 12$ .

|          |                    |                                      |
|----------|--------------------|--------------------------------------|
| <b>A</b> | <b>Correct</b>     | $24 \div 4 = 6$ .                    |
| B        | Dropped element    | $12 \div 4$ forgets the factor of 2. |
| C        | Wrong stem number  | The edge count.                      |
| D        | Inverted operation | Multiplies $12 \times 4$ .           |

### 12. Answer B Concept 8

A school district with 3 campuses connects 25 computers so that every pair can communicate, with no redundant loops. How many cables are needed?

**Target word:** edges in a tree **Decides:** Connected and acyclic means a tree:  $n - 1$ .

|          |                   |                                         |
|----------|-------------------|-----------------------------------------|
| A        | Confusion set     | Complete graph, $25 \times 24 \div 2$ . |
| <b>B</b> | <b>Correct</b>    | $25 - 1 = 24$ .                         |
| C        | Wrong stem number | One per computer adds a loop.           |
| D        | Camouflage        | Campus count does not decide edges.     |

## Triage grid

Write each item number in the box it belongs to.

|                       | Right                                                 | Wrong                                                                                 |
|-----------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Sure</b>           | Owned.                                                | <b>Concept inversion — top priority.</b> A wrong mental model; it survives rewording. |
| <b>Unsure / guess</b> | Recognition, not mastery. Re-drill with the transfer. | Not learned yet.                                                                      |

## Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

| Trap               | What it does                                                 | In this drill | Your misses |
|--------------------|--------------------------------------------------------------|---------------|-------------|
| Dropped element    | Leaves out a required term or step.                          | 9             |             |
| Wrong stem number  | A figure copied from the prompt that is not the answer.      | 3             |             |
| Adjacent category  | The right idea in the wrong bucket.                          | 6             |             |
| Inverted operation | Direction flipped: add/subtract, cause/effect, higher/lower. | 6             |             |
| Absolute language  | Always, never, guarantee — where judgment applies.           | 2             |             |
| Confusion set      | Every option is real; the test is exact role-matching.       | 8             |             |
| Camouflage         | Answers a stem fact that was never the question.             | 2             |             |

# Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

| # | Concept                                    | Rule                                                                                                                                                                 | Trap                                                       | Tell                                                                  |
|---|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|-----------------------------------------------------------------------|
| 1 | <b>Conditional statements</b>              | $p \rightarrow q$ is false only when $p$ is true and $q$ is false; it is equivalent to its contrapositive $\neg q \rightarrow \neg p$ , not its converse or inverse. | Treating the converse as equivalent.                       | An “if ... then ...” policy and “which is equivalent?”                |
| 2 | <b>Negating quantifiers</b>                | $\neg \forall x P(x) \equiv \exists x \neg P(x)$ and $\neg \exists x P(x) \equiv \forall x \neg P(x)$ .                                                              | Negating “all did” as “none did.”                          | “Every” or “some” in a statement to be negated.                       |
| 3 | <b>Inclusion-exclusion</b>                 | $ A \cup B  =  A  +  B  -  A \cap B $ .                                                                                                                              | Adding the sets without removing the overlap.              | Two overlapping groups with a “both” count.                           |
| 4 | <b>De Morgan’s laws in Boolean algebra</b> | $(xy)' = x' + y'$ and $(x + y)' = x'y'$ : complement each variable and swap AND with OR.                                                                             | Complementing the variables but keeping the same operator. | A complement bar over a whole product or sum.                         |
| 5 | <b>Arithmetic and geometric series</b>     | Arithmetic sum $= n/2 \times (\text{first} + \text{last})$ ; geometric sum $= a(r^n - 1)/(r - 1)$ .                                                                  | Reporting the last term instead of the sum.                | A pattern that grows by a fixed amount or a fixed ratio, and “total.” |
| 6 | <b>Relation properties</b>                 | An equivalence relation is reflexive, symmetric, and transitive; a partial order is reflexive, antisymmetric, and transitive.                                        | Calling any “same as” relation a partial order.            | “Is related to” defined by sharing a trait vs. by ordering.           |
| 7 | <b>Handshaking lemma</b>                   | The sum of all vertex degrees equals twice the number of edges.                                                                                                      | Forgetting that each edge is counted at both ends.         | Vertices with given degrees and a question about edges.               |
| 8 | <b>Trees</b>                               | A tree with $n$ vertices is connected, has no cycles, and has exactly $n - 1$ edges.                                                                                 | Using $n$ edges or the complete-graph count.               | “Connected with no loops” or “no redundant links.”                    |

**Course scope (public description):** Discrete Mathematics I helps candidates develop competence in the use of abstract, discrete structures fundamental to computer science. In particular, this course will introduce candidates to logic and proofs; Boolean algebra and functions; set theory; finite and infinite sequences and series; and relations, graphs, and trees. The course emphasizes applications in computer science. Calculus I is a prerequisite for this course.